

Problem statement 1:

Problem Statement:

Given a positive integer N , print all the numbers from 1 to N in increasing order, separated by a single space.

Your task is to write a program that prints the required sequence.

After completing the program, analyze your solution by calculating its Time Complexity and Space Complexity, and explain why your answer is correct.

Input Format

A single integer N .

Output Format

Print all integers from 1 to N in increasing order separated by spaces.

Constraints

$1 \leq N \leq 10^5$

Example 1

Input

5

Output

1 2 3 4 5

Explanation

The numbers from 1 to 5 are printed in increasing order.

Example 2

Input

8

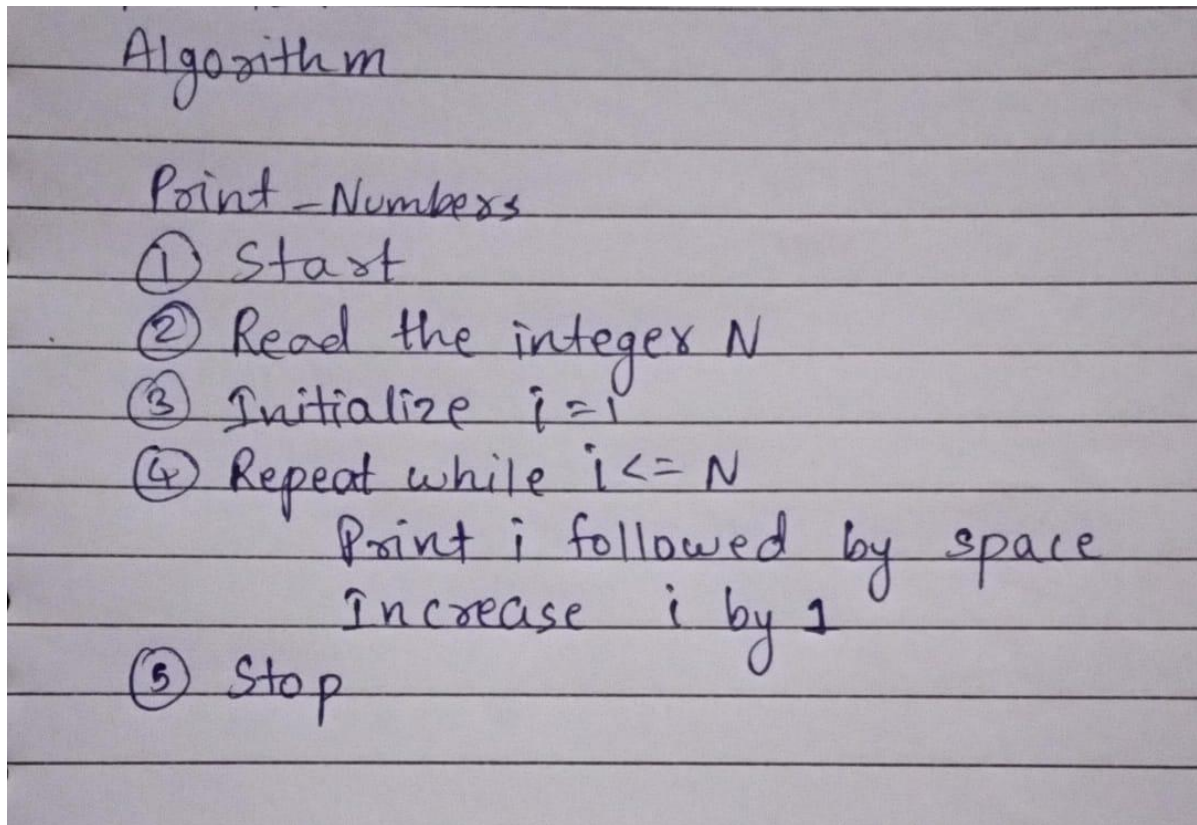
Output

1 2 3 4 5 6 7 8

Explanation

The program prints every integer starting from 1 until 8, maintaining increasing order.

Algorithm:



Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int N;
7      std::cin >> N;
8
9      for (int i = 1; i <= N; i++)
10     {
11         cout << i << " ";
12     }
13
14     return 0;
15 }
16
```

Output:

```
PS C:\Users\priya\OneDrive\Desktop\Week 4 Module> .\Problem_1.exe
5
1 2 3 4 5
```

Complexity:

Complexity

Time —
 $O(N)$

- The loop executes exactly ~~N~~ N times, once for each number from 1 to N

Space —
 $O(1)$

- The program only uses a fixed number of variables (N and i). The amount of memory used does not ~~represent~~ increase with N

Problem statement 2:

Problem Statement:

Given a positive integer N, find the sum of the first N natural numbers using a loop.

Your task is to write a program that calculates and prints the required sum.

After completing the program, analyze your solution by calculating its Time Complexity and Space Complexity, and explain your answer.

Input Format

A single integer N.

Output Format

Print the sum of the first N natural numbers.

Constraints

$$1 \leq N \leq 10^7$$

Example 1

Input

5

Output

15

Explanation

The sum of the first five natural numbers is:

$$1 + 2 + 3 + 4 + 5 = 15$$

Example 2

Input

10

Output

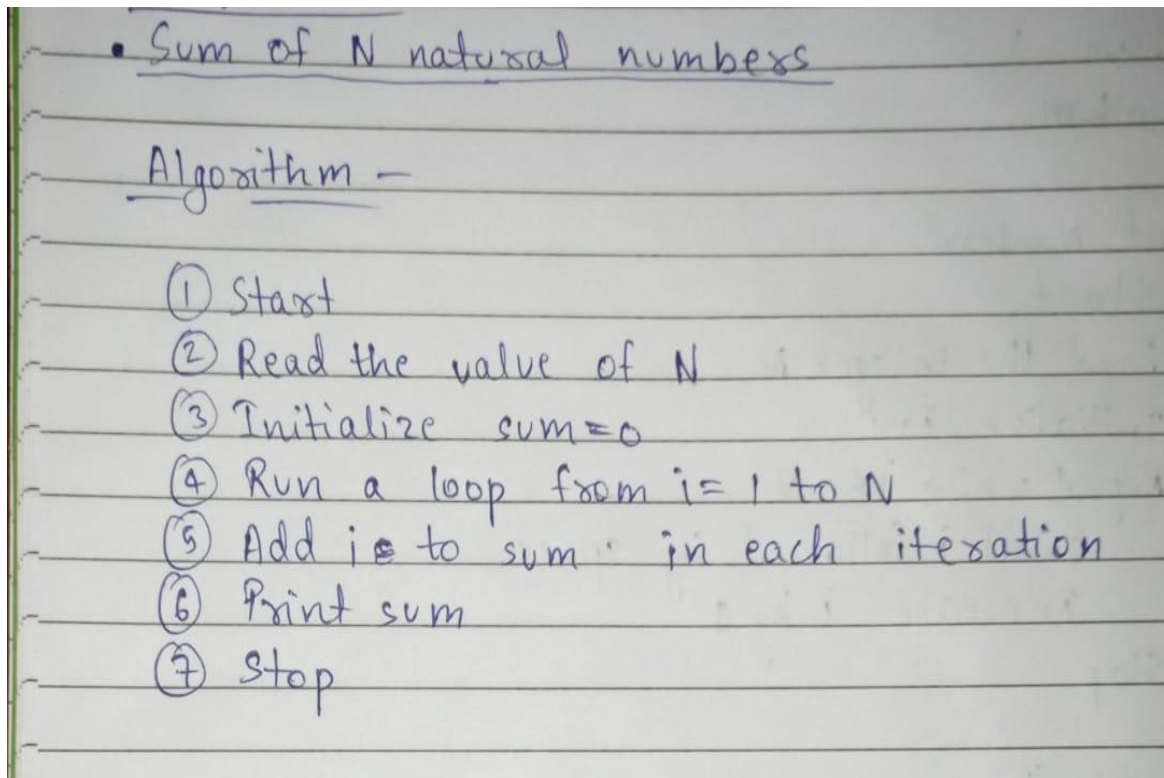
55

Explanation

The sum of the first ten natural numbers is:

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

Algorithm:



Code:

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int N;
7      long long sum = 0;
8
9      cin >> N;
10
11     for (int i = 1; i <= N; i++)
12     {
13         sum = sum + i;
14     }
15
16     cout << sum << endl;
17
18     return 0;
19 }
```

Output:

```
PS C:\User
5
15
PS C:\User
8
36
PS C:\User
```

Complexity:

Complexity —

- Time — $O(N)$
Execution increases with N
as loop repeats itself N times.
- Space — $O(1)$
Fixed no. of variables (N, i, sum) are
used regardless of N 's value